4.2 Complex Solutions of Equations

Name _____ Period ____ Date _____

Pre-Calculus 4.2 · Student sheet

I can…

- tell how many solutions a polynomial equation has.
- use the discriminant to describe the solutions of a quadratic equation.
- find all the complex zeros of a polynomial.

Vocabulary · fill in the missing word

Fundamental Theorem of Algebra — every _____ of degree $n > 0$ has at least one zero in the complex number system

Discriminant — $b^2 - 4ac$, the _____ under the radical in the Quadratic Formula

Linear Factorization Theorem — a _____ of degree n has exactly n linear factors, $f(x) = a(x - c_1)(x - c_2)\cdots(x - c_n)$

Part 1 · Graphs and discriminants

Graph each one in Desmos. Count the x-intercepts and find $b^2 - 4ac$.

Equation	x-intercepts	$b^2 - 4ac$
$y = x^2 - 4x + 3$		
$y = x^2 - 4x + 4$		
$y = x^2 - 4x + 5$		

Part 2 · Notes and examples (fill in each note, then work the example)

Number of solutions: a polynomial equation of degree n has exactly _____ solutions in the complex numbers (counting repeated ones).

Example 1

How many solutions does $x^3 + 4x = 0$ have? Find them.

Discriminant $b^2 - 4ac$: positive → two real solutions; zero → one repeated real solution; negative → two _____.

Example 2

Describe the solutions.

a. $x^2 + 2x + 5 = 0$

b. $4x^2 - 12x + 9 = 0$

Example 3

Solve $x^2 + 2x + 5 = 0$.

Complex Conjugate Zeros: if $a + bi$ ($b \neq 0$) is a zero of a polynomial with real coefficients, then _____ is also a zero.

Example 4

$1 + 3i$ is a zero of $f(x) = x^4 - 3x^3 + 6x^2 + 2x - 60$. Find all the zeros.

Example 5

Write $x^4 - 16$ as a product of linear factors.

Part 3 · Partner practice

Partner 1

Solve $x^2 - 4x + 13 = 0$.

Partner 2

Find all the zeros of $f(x) = x^3 - x^2 + 4x - 4$.